

Session 2: Block Data & Drawing a piece

```
#include <stdlib.h>
```

```
#include <time.h>
```

랜덤 함수 사용하기 위해 헤더 파일 추가

```
const int shapes[7][4][4][4] = {...
```

shapes 리스트에 다른 모양들 추가

```
void InitializeGame() {
    // To-do: Initialize the game board to be empty (0).
    // Hint: Game board is essentially 2D array with ROWS and COLUMNS.
    // Hint: int board[ROWS][COLUMNS];
    for (int row = 0; row < ROWS; row++) {
        for (int col = 0; col < COLUMNS; col++) {
            board[row][col] = 0;
        }
    }

    // To-do: Initialize the first tetromino block using the global variables
    defined earlier.
    srand(time(NULL));
    currentPieceX = 3;
    currentPieceY = 0;
    currentPieceRotation = rand() % 4;
    currentPieceType = rand() % 7;

    printf("Game Initialized. Current piece type: %d at board coordinates
    (%d, %d), rotation: %d\n",
           currentPieceType, currentPieceX, currentPieceY, currentPieceRotation);
}
```

InitializeGame 함수에 보드 비우는 것과 첫 테트로미노 블록 초기화 기능 추가

```
void DrawCurrentPiece() {
    // Function to draw the current Tetromino piece on the board.

    // To-do: Calculate which block in the board[ROWS][COLUMNS] should be
    filled based on the "shapes[7][4][4][4]" array
    // and the global variables currentPieceX, currentPieceY,
    currentPieceType, and currentPeiceRotation
```

```

// that you initialized in InitalizeGame() function.
int tetrominos[4][2]; // [row, col]
int num = 0;
for (int i = 0; i < 4; i++) {
    for (int j = 0; j < 4; j++) {
        if (shapes[currentPieceType][currentPieceRotation][i][j] == 1) {
            tetrominos[num][0] = i + currentPieceY;
            tetrominos[num++][1] = j + currentPieceX;
        }
    }
}

// To-do: Calculate the actual pixel position of the block using
BOARD_OFFSET and BLOCK_SIZE.
// the pixel position of the block [X][Y] is [BOARD_OFFSET_X +
X*BLOCK_SIZE][BOARD_OFFSET_Y + Y*BLOCK_SIZE]
int tetromino_pixels[4][2]; // [x, y]
for (int i = 0; i < 4; i++) {
    tetromino_pixels[i][0] = tetrominos[i][1] * BLOCK_SIZE +
BOARD_OFFSET_X;
    tetromino_pixels[i][1] = tetrominos[i][0] * BLOCK_SIZE +
BOARD_OFFSET_Y;
}

// To-do: Use DrawBlock function below to draw a block starting at
(pixelX, pixelY) pixel coordinate.
for (int i = 0; i < 4; i++) {
    DrawBlock(tetromino_pixels[i][0], tetromino_pixels[i][1]);
}
}

```

DrawCurrentPiece 함수 수정

tetrominos : board에서의 테트로미노의 위치 (row, col)

tetromino_pixels : 픽셀 위치로의 테트로미노의 위치 (x, y)

(프로그램 화면에서 y축이 반대 방향인 것에 유의)

각 블록마다 DrawBlock함수로 게임 창에 그리기